AI-POWERED PRICING

Agentic AI-Based Dynamic Tariff Optimization for EV Charging Networks

Using UrbanEV and ACN Charging Session Data

Presented by:

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Executive Summary

OBJECTIVE

- Develop an Agentic AI framework that predicts charging demand, identifies congestion, and dynamically adjusts EV charging tariffs.
- Improve network revenue and infrastructure utilization through real-time, data-driven pricing.

APPROACH

- UrbanEV: behavioral insights; ACN: model training, validation & pricing evaluation.
- Four sequential agents: Real-Time Demand Prediction → Congestion Monitoring → Dynamic Pricing → Monitoring & Feedback.

KEY RESULTS

- Demand Prediction (XGBoost): R^2 = 0.811
- Congestion Agent: Accuracy = 91%, F1 = 0.59
- Est. revenue uplift: +12.33% (fixed-demand assumptions).
- Framework demonstrates real-time data can increase revenue & visibility.

UrbanEV was used for behavioral insights, while **ACN** was used for model training, validation, and pricing evaluation.

Limitations

- Dynamic pricing evaluated through simulation.
- Elasticity not directly estimated from historical data.
- ACN primarily reflects workplace charging behaviour.

Data Landscape & Problem Understanding

UrbanEV Dataset

Behavioral Insight Dataset

247 stations **30** days **2.13M** records

Purpose: Understand large-scale charging demand, congestion dynamics, and pricing behavior.

ACN Dataset

Operational Modeling Dataset

14,999 sessions

Session-level charging behavior

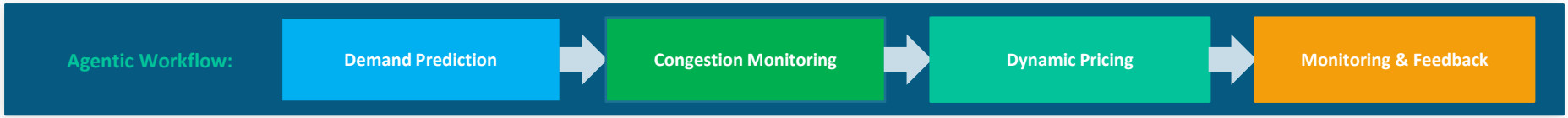
Energy delivered, duration, station info

Purpose: Develop the operational real-time demand prediction and dynamic pricing system.

UrbanEV was used for behavioral insights, while **ACN** was used for model training, validation, and pricing evaluation.

Key Challenge

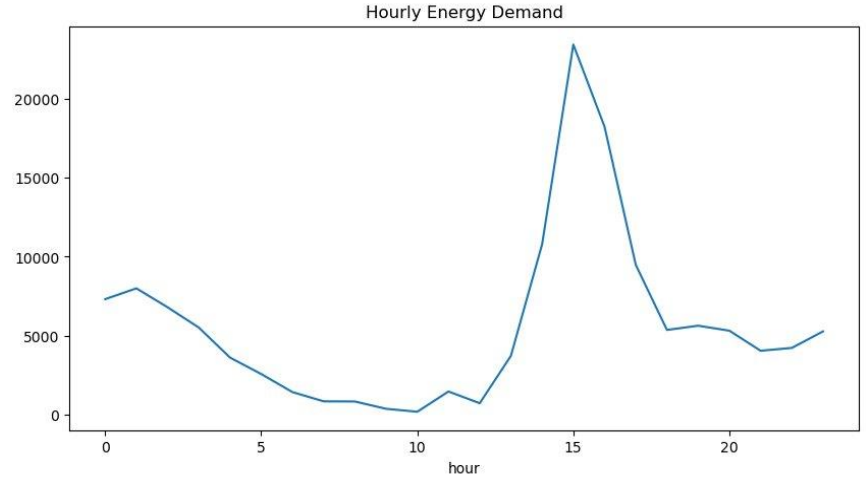
Most charging networks use static tariffs despite significant demand variation — causing peak-hour congestion, underutilized off-peak infrastructure, and suboptimal revenue.



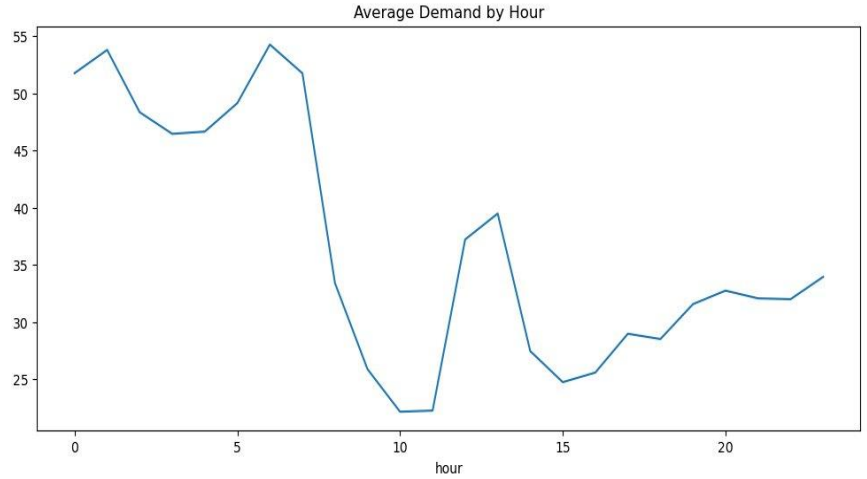
Specialized agents collaborate through a continuous decision-support workflow.

Charging Demand & User Behaviour Insights

ACN Hourly Demand Profile



UrbanEV Hourly Demand Profile

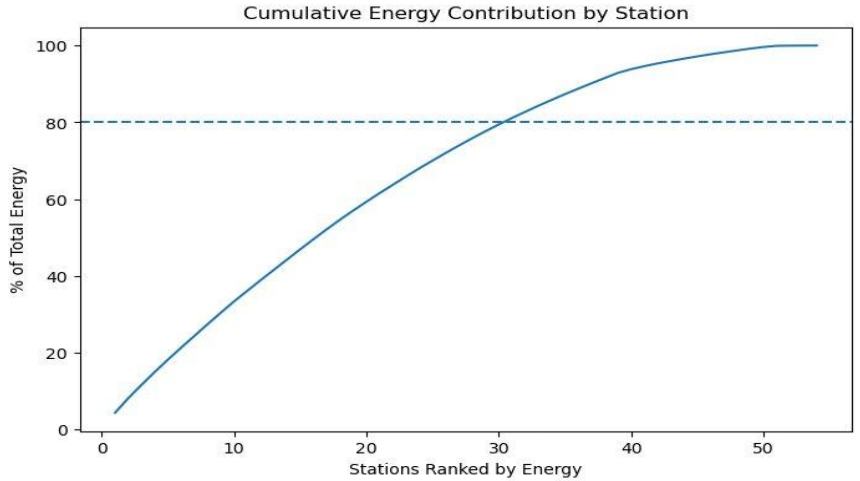


Key Findings

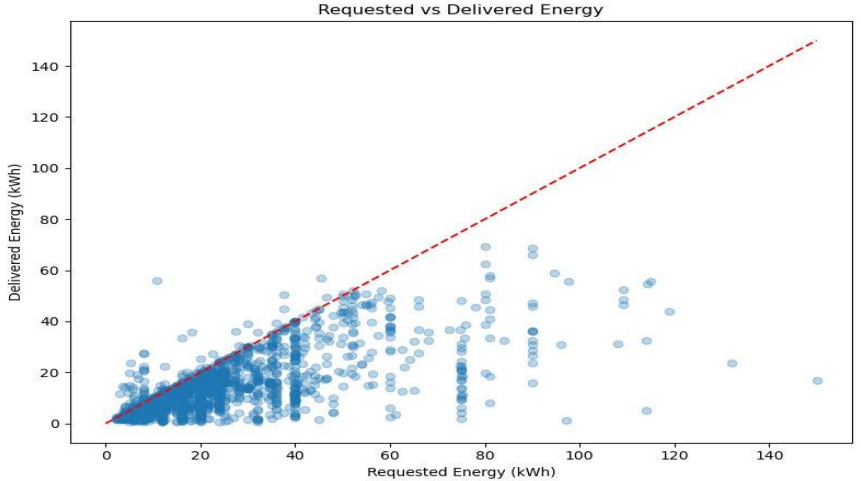
- Both datasets exhibit strong temporal demand variation.
- UrbanEV demonstrates city-scale charging demand cycles.
- ACN demonstrates workplace charging behavior, with peak activity during afternoon hours.
- Weekday demand consistently exceeds weekend demand.
- These observations indicate that static pricing fails to reflect actual charging demand conditions.

Infrastructure Utilization Analysis

Cumulative Energy Contribution by Station



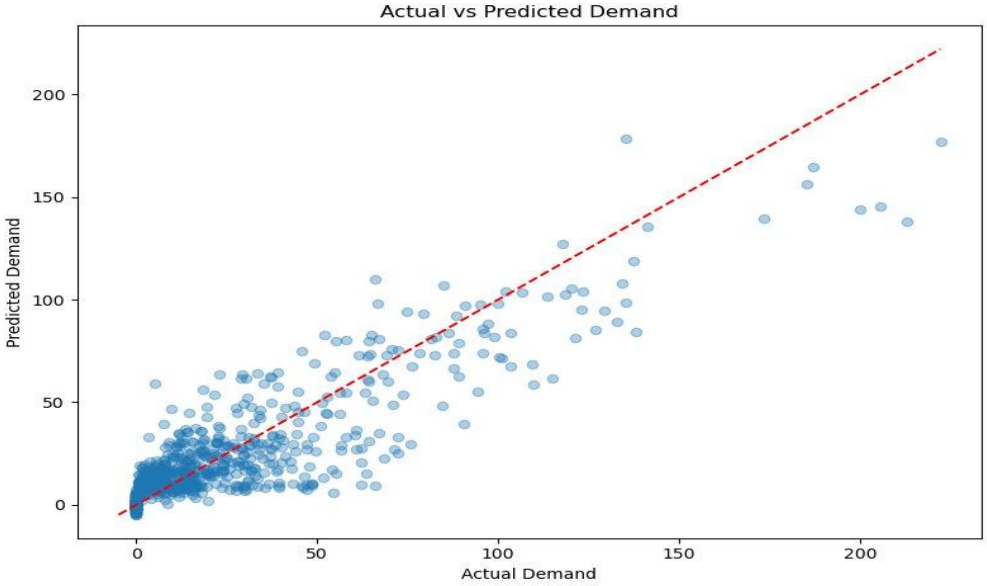
Requested vs Delivered Energy



Key Findings

- ~30 stations account for 80% of total energy delivered — demand is highly concentrated.
- Average request fulfillment is approximately 65%, indicating users frequently receive less energy than requested.
- Vehicles often remain connected after charging completion, reducing charger availability.
- Station utilization varies substantially across the network, creating opportunities for congestion-aware pricing strategies.

Real-Time Demand Prediction Agent



Model	R^2
Naive Baseline	0.105
Historical Features Only	0.395
Real-Time Agent (XGBoost)	**0.811**

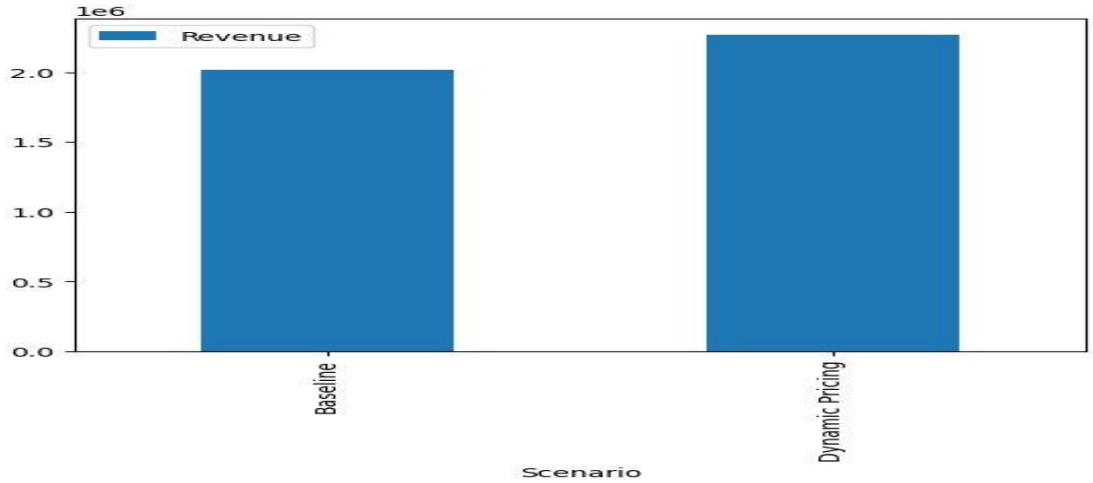
Real-time charging activity significantly improves demand prediction accuracy compared with historical-only approaches.

Why XGBoost?

- Handles nonlinear charging demand patterns.
- Robust to noisy operational data.
- Captures time, demand & session-count interactions.
- Consistently outperformed all baseline approaches.

Real-Time Operational System: This agent operates as a real-time decision-support system using live session data — not a long-horizon forecasting model. Predictions are generated per charging cycle to feed the pricing engine immediately.

Dynamic Tariff Pricing Agent



Pricing Logic

Baseline Tariff: ₹15/kWh

Peak Demand	+15% → ₹17.25/kWh
High Demand	+8% → ₹16.20/kWh
Low Demand	−10% → ₹13.50/kWh

Tariff (₹/kWh) adjusts in real-time based on predicted demand levels.

Pricing Pipeline:

Demand Prediction → Demand Classification → Tariff Adjustment → Revenue Simulation

Baseline Revenue

₹ 2.02 Million

Dynamic Revenue

₹ 2.27 Million

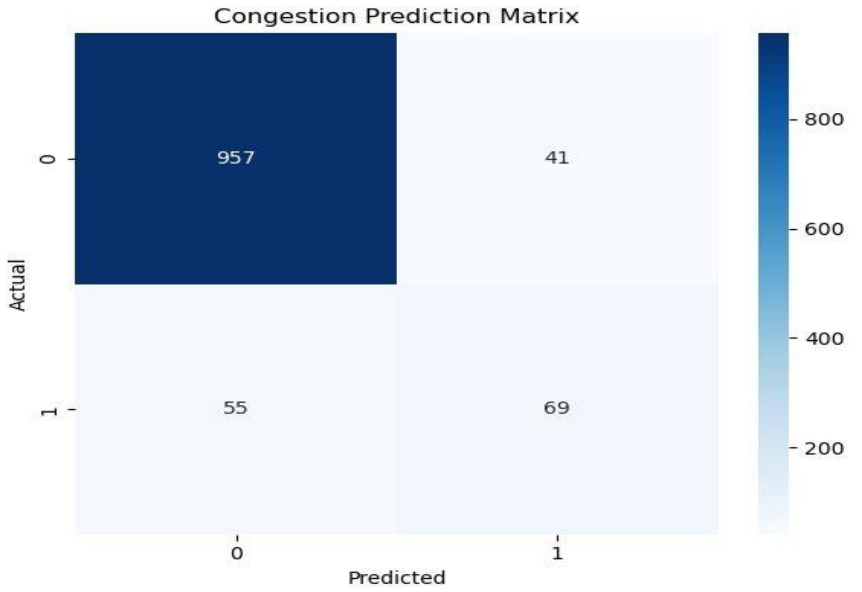
Est. Uplift (fixed-demand)

+12.33%

Elasticity: Sensitivity testing showed revenue gains remained positive under multiple customer-response assumptions. See Appendix D.

Congestion & Monitoring Agent

Congestion Definition: Session Count > Historical 80th Percentile · *80th percentile selected to balance operational sensitivity and training-data availability.*



Monitoring Agent Responsibilities

- Monitor real-time demand forecasts from the Prediction Agent.
- Track congestion probability.
- Evaluate revenue performance.
- Monitor charging utilization.
- Provide feedback for future tariff adjustments.

Performance Metrics

91%	63%	56%	59%
Accuracy	Precision	Recall	F1 Score

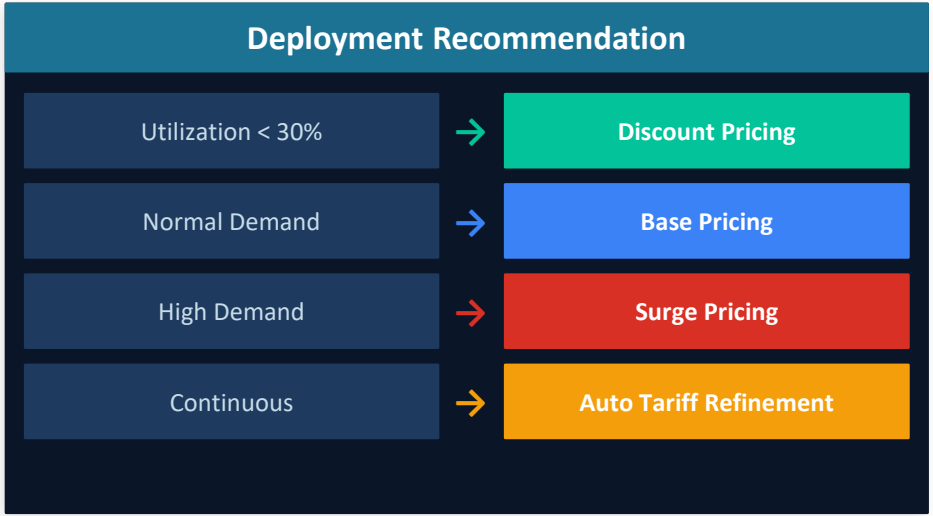
Congestion events are relatively infrequent; F1 score provides a more informative assessment than accuracy alone.

Business Impact & Recommendations

Business Impact

- Estimated revenue uplift of 12.33% under fixed-demand assumptions.
- Real-time demand prediction accuracy improved significantly vs. naive baseline methods.
- Congestion periods can be identified proactively rather than reactively.
- Charging infrastructure utilization can be managed more effectively through demand-responsive pricing.

Deployment Recommendation



Conclusion

The proposed Agentic AI framework operates as a real-time operational decision-support system — not a long-horizon forecasting tool. It dynamically optimizes EV charging tariffs using live session data, producing an estimated 12.33% revenue uplift under fixed-demand assumptions.

Appendix A Why XGBoost?

Alternative Models Considered

Linear Regression

Unable to capture nonlinear demand patterns.

Decision Trees

High variance and overfitting risk.

Random Forest

Strong performance but weaker predictive performance and interpretability compared with the Real-Time Demand Prediction Agent (XGBoost).

Selected Model: XGBoost

Reasons

- Superior predictive performance.
- Robust handling of missing and noisy operational data.
- Effective modeling of nonlinear charging demand behavior.
- Widely adopted in demand forecasting and energy analytics applications.

Appendix B Model Validation & Leakage Checks

Validation Checks Performed

- Time-based train-test split used.
- No random shuffling applied.
- Naive baseline comparison performed.
- Historical-only model comparison performed.

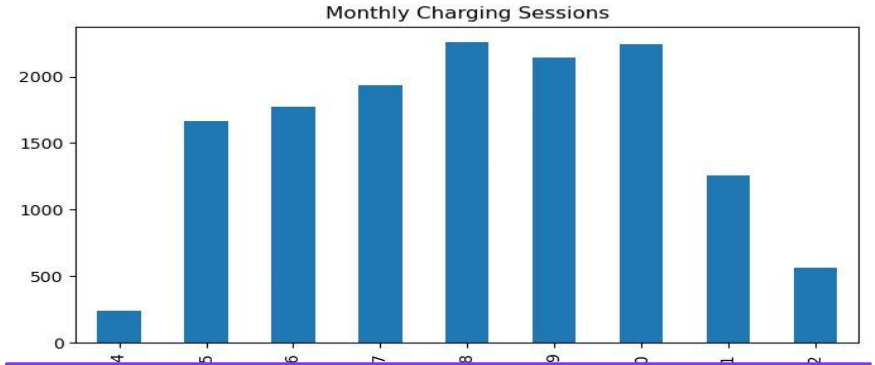
Results

Naive Baseline	$R^2 = 0.105$
Historical Features Only	$R^2 = 0.395$
Real-Time Prediction Agent	$R^2 = 0.811$

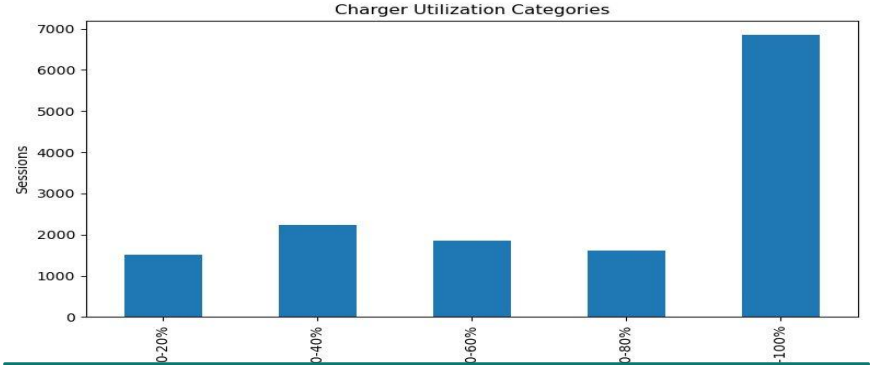
These results indicate that the model learns meaningful demand patterns beyond simple historical averages while benefiting from real-time operational information.

Appendix C Additional Insights

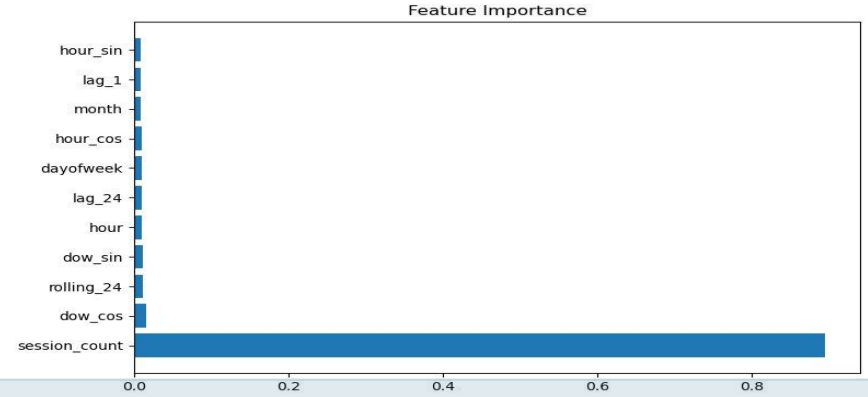
Monthly Charging Sessions



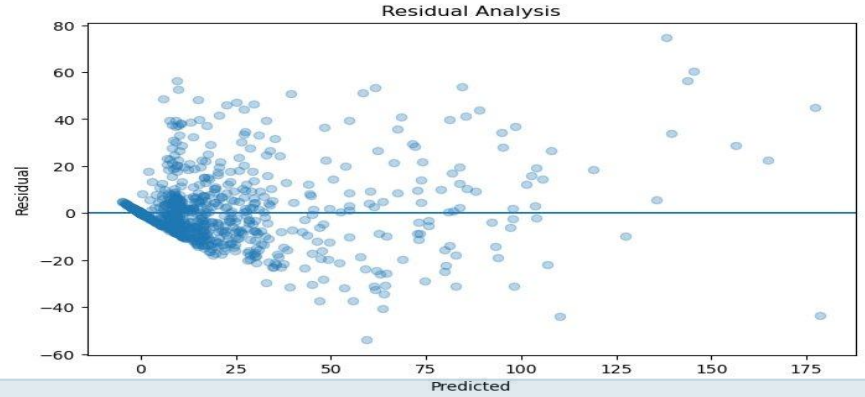
Charger Utilization Categories



Feature Importance



Residual Analysis



Additional EDA confirms strong temporal charging behavior, seasonal demand variation, and heterogeneous station utilization patterns across the network.

Appendix D Elasticity Robustness Check

Why Sensitivity Analysis?

- Limited historical tariff variation in the ACN dataset prevented direct elasticity estimation.
- Customers were always charged the same rate historically, so their price-response behaviour is unobserved.
- Sensitivity testing was performed instead, simulating revenue across a plausible elasticity range.
- This approach follows standard practice in demand-response feasibility studies.

Sensitivity Results

Elasticity Range Tested: −0.1 to −0.3

ε = −0.1 (low sensitivity)	**Revenue uplift retained**
ε = −0.2 (moderate)	**Revenue uplift retained**
ε = −0.3 (high sensitivity)	**Revenue uplift retained**

Revenue remained positive across all tested elasticity scenarios, supporting the feasibility of dynamic pricing deployment.

Note: Elasticity was not directly learned due to limited historical tariff variation. Sensitivity testing across elasticity scenarios (−0.1 to −0.3) continued to show positive revenue uplift, supporting pricing robustness.